

TM Forum Component

Fault Management

TMFC043

Maturity Level: General Availability (GA)	Team Approved Date: 2026-05-21
Release Status: Production	Approval Status: TM Forum Approved
Version 1.1.0	IPR Mode: RAND

Notice

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TMFC043 – Fault Management

1. Overview

Component Name	ID	Description	ODA Function Block
Fault Management	TMFC043	Fault Management component involves identifying, diagnosing, and resolving issues that prevent the communication resources and their overlaying services from functioning correctly. Its scope covers this functionality at the ODA Production layer while business related resolutions are not part of its scope. The Fault Management component manages the currently active alarms or problem candidates, potentially modifying or enhancing their data to make persistent alarms more actionable. It may apply logic to preserve alarms that reflect actual problems, eliminating those that are merely symptoms. Various actions can be taken using the alarm/problem information as part of the resolution process, such as applying automated/autonomous corrective actions or opening trouble tickets.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
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1.5.8	L2	Resource Trouble Management	Resource Trouble Management processes are responsible for the management of troubles, including security events, associated with specific resources. The objectives of these processes are to efficiently and effectively manage reported resource trouble, isolate the root cause and act to resolve the resource trouble.
1.4.6	L2	Service Problem Management	Service Problem Management processes are responsible for the management of problems associated with specific services. The objective of these processes is to respond immediately to reported service problems or failures in order to minimize their effects on customers, and to invoke the restoration of the service, or provide an alternate service as soon as possible.

2.2. SID ABEs

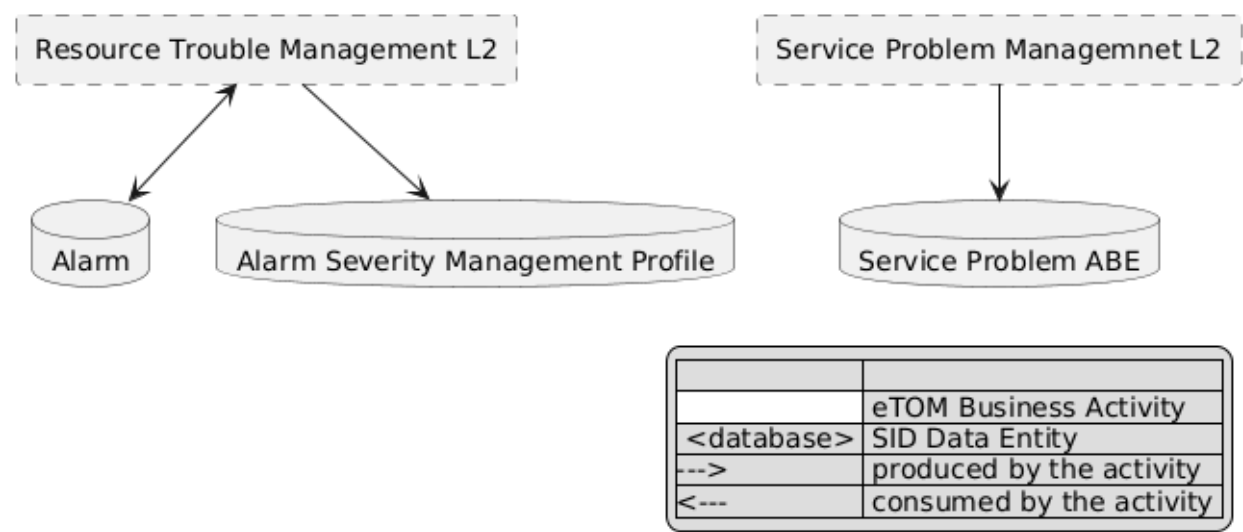
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)	SID ABE L1 Definition	SID ABE L2 Definition
Resource Trouble	Resource Alarm	The Resource Trouble ABE collects information about problems found in allocated resource instances, regardless of whether the problem is physical or logical. Entities in this ABE capture detection, root cause, and resolution of these problems and maintain a history of the activities involved in diagnosing and solving the problem. This history includes tracking, reporting, assigning people to fix the problem, testing and verification, and overall administration of repair activities.	<i>(no description available)</i>

Resource Trouble	AlarmSeverityAssignmentProfile	The Resource Trouble ABE collects information about problems found in allocated resource instances, regardless of whether the problem is physical or logical. Entities in this ABE capture detection, root cause, and resolution of these problems and maintain a history of the activities involved in diagnosing and solving the problem. This history includes tracking, reporting, assigning people to fix the problem, testing and verification, and overall administration of repair activities.	This object class models the (flexible) severity assignment to specified probable causes. So the ASAP object includes a table, with each row specifying the probable cause (plus optionally the probableCauseQualifier and / or the nativeProbableCause) and the assigned severity for "service affecting", "non service affecting" and "service independent or unknown" alarms. Probable cause, probableCauseQualifier and nativeProbableCause are the coordinates of the ASAP entries, so there must not be two ASAP entries with the same coordinates. It is the responsibility of the EMS to enforce this. Alarms against entities that are not modeled by the interface are reported using the "AID" objectType. For these "AID" alarms the ASAP mechanism is not applicable. Typically the ASAP function is implemented in the OS or NE where the alarm is originated. I. e. the severity will usually be assigned directly by the NE, so it is up to the EMS to download the defined ASAP to the NE. Other alarms, e.g. on arcs like SNCs, may be generated by the target OS. An alarm becomes reportable by NE/target OS: - when for PTP, CTP, FTP: AlarmReporting changes from "off" to "on" - when for SNC, TopologicalLink, Equipment, EquipmentHolder, GTP: alarmReportingIndicator changes from false to true. Alarms are always reportable for all other objects as they do not have any alarm reporting attribute. When an alarm becomes reportable the ASAPs are applied (accessed via the asapPointer). The severity of the alarm is adopted from the ASAP entry that matches for all of the following: - same probableCause - same probableCauseQualifier (an empty string is a match) - same nativeProbableCause (an empty string is a match). For example, if the reportable alarm has LOS probableCause and an ASAP entry is found with LOS probableCause and both probableCauseQualifier and nativeProbableCause are empty strings, then the search is successful. If there is a matching ASAP, then the severities are adopted on the following basis: - the alarm is service affecting: the severity specified in the serviceAffecting field is assigned - the alarm is service non affecting: the severity specified in the serviceNonAffecting field is assigned - the alarm is service independent or NE/target OS is not equipped to determine if the alarm affects service or not: the severity specified in the serviceIndependentOrUnknown field is assigned. If the assigned severity specified in the ASAP is: - "FREE_CHOICE" then the NE/target OS are free to determine the severity. If there is no local mechanism on the ME/EMS to determine severity then the severity "INDETERMINATE" should be assigned. - "NOTALARMED" then the EMS should not emit an alarm over the EML-NML interface. If there is no ASAP that matches, then the NE/target OS is free to determine the severity. If there is no local mechanism on the NE/target OS to determine severity then the severity "INDETERMINATE" should be assigned. Once a severity has been assigned, the alarm notification is emitted (other than for "NOTALARMED"). Note: Any operation of alarm retrieval will not include "NOTALARMED" alarms.
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Service Problem		The Service Problem ABE manages faults, alarms, and outages from a Service point-of-view. This is then correlated to trouble tickets, regardless of whether the cause is physical or logical. Other entities in this ABE are used to direct the recovery from each of these types of problems. They provide the ability to associate Resource faults and alarms to degradation and outages of Services that run on those Resources.	(no description available)
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2.3. eTOM L2 - SID ABEs links



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2

1187	Resource Trouble Resolution	Resource Trouble Resolution Function identifies the necessary resolution actions to solve the Resource Trouble according to Resource Problem Policy. It is based on diagnostic results.	Resource Trouble Management	Resource Trouble Analysis
615	Service Problem Tracking & Management	Service Problem Tracking & Management functions provides necessary functionality to assure that service problems are assigned, coordinated, and restored efficiently, escalating as needed.	Service Problem Management	Service Problem Repository Management
1155	Resource Diagnostic and Test Analysis	Resource Diagnostic and Test Analysis Function includes: - Management of success/failure result feedback of diagnostic and correction provided by technicians or the systems after a correction - Offline analysis of the success/failure data compared to the diagnostic and correction rules to provide modifications to improve them. This analysis is done using statistical diagnostic data and test results to evaluate their relevance and the needed updates.	Resource Strategic Analysis	Resource Diagnostic and Test Quality Loop

1186	Resource Trouble Cause Identification	Resource Trouble Cause Identification Function supports the trouble causes identification, that is to say the reasons why the installed resource is in trouble and among these reasons the root cause of the resource trouble. It implements Resource Problem Policy composed of diagnostic actions (Test, inventories and historic consultation, workforce intervention, etc.) The identification of the resource trouble cause can be made through: - Analysis of associated Service Trouble Ticket Information (first diagnostic) - Analysis of associated Resource Trouble Ticket Information (first diagnostic) - Analysis of Resource alarm events (fault of performance data) - Issuing resource tests, including one time tests or a series of tests over a period of time - Analysis of previous resource test results, resource usage historic, QoS historic, equipment logs, anomaly historic, etc.	Resource Trouble Management	Resource Trouble Analysis
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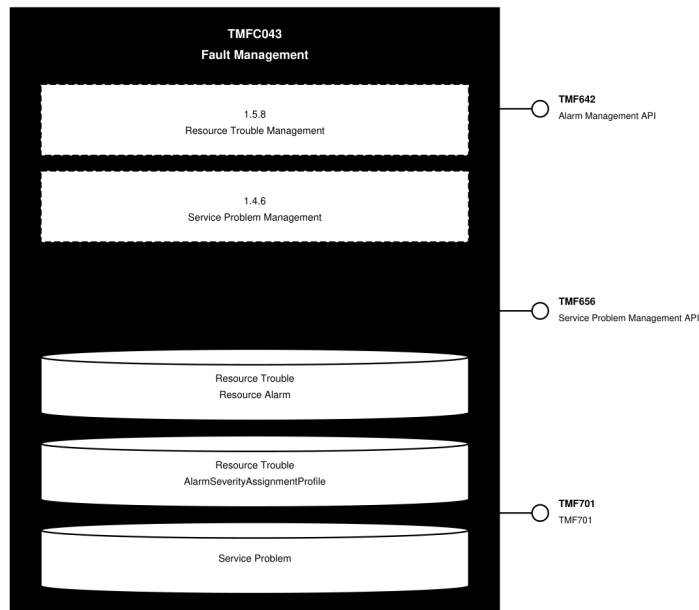
610	Service Problem Prioritization	Prioritization of currently open service problems, giving consideration to problem impact, service level agreement, etc. The function will send notifications to operations applications for manual or automatic mitigation or correction.	Service Problem Management	Service Problem Resolution
612	Service Problem Automatic Correction	Service Problem Automatic Correction functions provide the necessary maintenance functionality to automatically resolve service problems back to a normal operational state or mitigation of the service problems. This involve automated restoration activities.	Service Problem Management	Service Problem Resolution
979	Service Assurance Control Calculation	Service Assurance Control Calculation calculates corrective actions to prevent or restore performance and quality. The calculations are rule based and takes the service's dynamics into account to perform a closed loop control to hold the performance and quality at a decided level.	Service Problem Management	Service Problem Resolution

980	Service Assurance Trouble Mitigation	Service Assurance Trouble Mitigation have autonomic rules to minimize the lack of service by doing mitigation action e.g. restart or re-routing etc. to mitigate the negative impact for the customer from a service problem or fall-out.	Service Problem Management	Service Problem Resolution
1067	Service Problem Manual Correction Support	Service Problem Manual Correction Support functions provide the functionality of dispatching of field technicians for manual restoration activities to resolve the service problem back to a normal operational state as efficiently as possible. This may involve reconfiguration, re-assignments, or temporary work-arounds	Service Problem Management	Service Problem Resolution

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

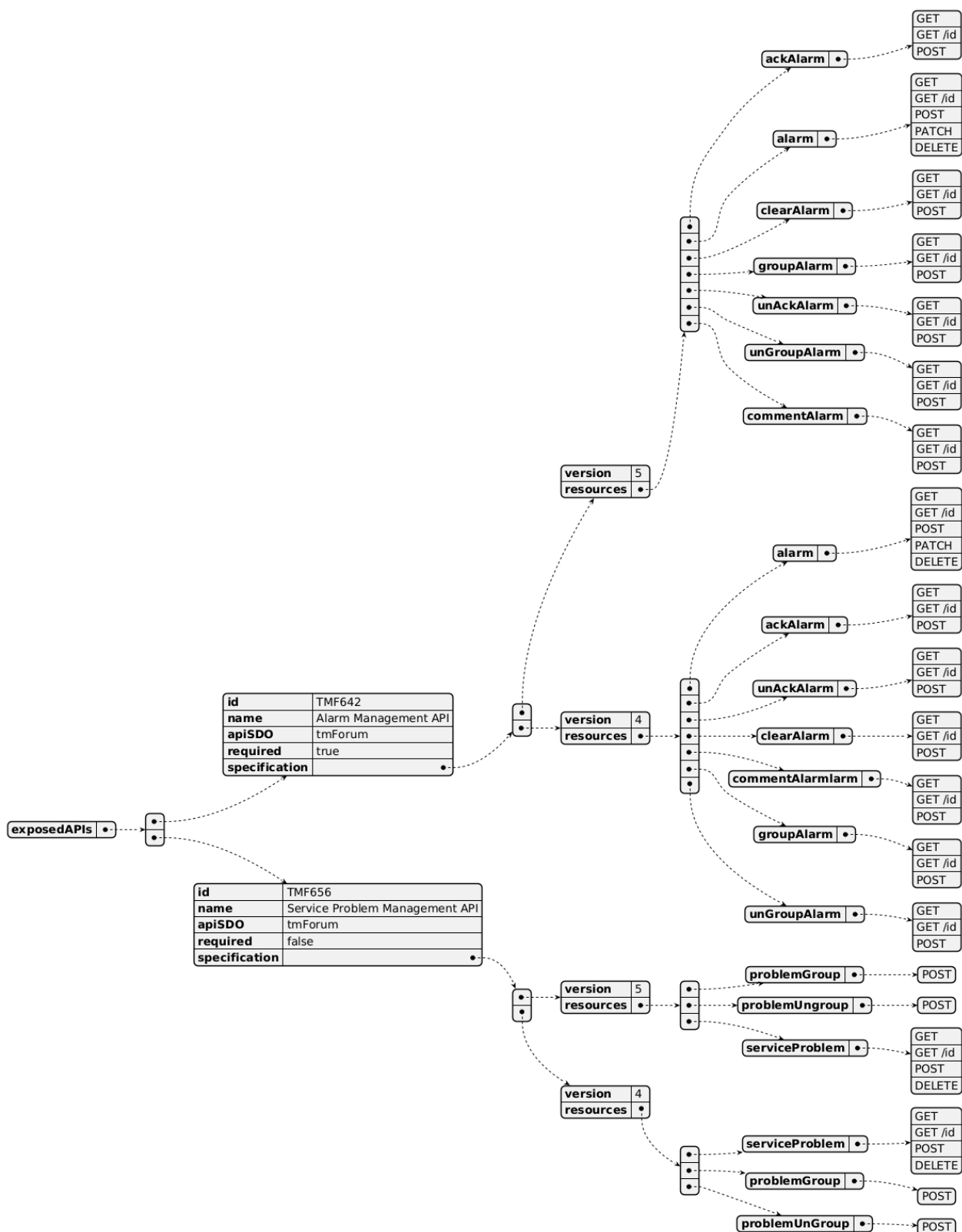


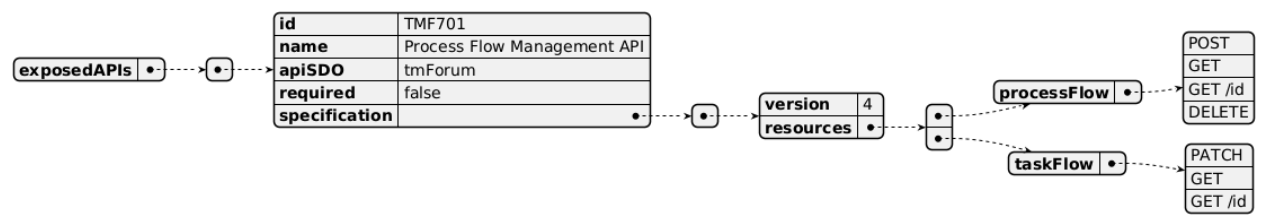
(SVG source: [TMFC043 API Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the *component-specification-documentation skill's scripts/render_api_context_svg.py*.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF642	Alarm Management API	Mandatory	5	ackAlarm alarm clearAlarm groupAlarm unAckAlarm unGroupAlarm commentAlarm	GET GET /id POST PATCH DELETE
TMF642	Alarm Management API	Mandatory	4	alarm ackAlarm unAckAlarm clearAlarm commentAlarm groupAlarm unGroupAlarm	GET GET /id POST PATCH DELETE

TMF656	Service Problem Management API	Optional	5	problemGroup problemUngroup serviceProblem	POST GET GET /id DELETE
TMF656	Service Problem Management API	Optional	4	serviceProblem problemGroup problemUnGroup	GET GET /id POST DELETE
TMF701	TMF701	Optional	4	processFlow taskFlow	POST GET GET /id DELETE PATCH





3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations



3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

API ID	API Name	Event Resources
TMF642	Alarm Management API	alarmCreateEvent alarmAttributeValueChangeEvent alarmStateChangeEvent alarmDeleteEvent ackAlarmsCreateEvent ackAlarmsStateChangeEvent unAckAlarmsCreateEvent unAckAlarmsStateChangeEvent clearAlarmsCreateEvent clearAlarmsStateChangeEvent commentAlarmsCreateEvent commentAlarmsStateChangeEvent groupAlarmsCreateEvent groupAlarmsStateChangeEvent unGroupAlarmsCreateEvent unGroupAlarmsStateChangeEvent

TMF701	ProcessFlowManagement	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent
TMF656	Service Problem Management API	problemAcknowledgementCreateEvent problemAcknowledgementDeleteEvent problemAcknowledgementStateChangeEvent problemGroupCreateEvent problemGroupDeleteEvent problemGroupStateChangeEvent problemUnacknowledgementCreateEvent problemUnacknowledgementDeleteEvent problemUnacknowledgementStateChangeEvent problemUngroupCreateEvent problemUngroupDeleteEvent problemUngroupStateChangeEvent serviceProblemAttributeValueChangeEvent serviceProblemCreateEvent serviceProblemDeleteEvent serviceProblemInformationRequiredEvent serviceProblemStateChangeEvent



Subscribed Events

API ID	API Name	Event Resources
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TMF633	Service Catalog Management API	serviceCatalogCreateEvent serviceCatalogChangeEvent serviceCatalogDeleteEvent serviceCategoryCreateEvent serviceCategoryChangeEvent serviceCategoryDeleteEvent serviceCandidateCreateEvent serviceCandidateChangeEvent serviceCandidateDeleteEvent serviceSpecificationCreateEvent serviceSpecificationChangeEvent serviceSpecificationDeleteEvent
TMF638	Service Inventory Management API	serviceCreateEvent serviceChangeEvent serviceDeleteEvent serviceStateChangeEvent
TMF639	Resource Inventory Management API	resourceCreateEvent resourceChangeEvent resourceDeleteEvent resourceStateChangeEvent
TMF673	Geographic Address Management API	geographicAddressValidationStateChangeEvent
TMF674	Geographic Site Management API	geographicSiteCreateEvent geographicSiteAttributeValueChangeEvent geographicSiteStatusChangeEvent geographicSiteDeleteEvent
TMF675	GeographicLocationManagement	geographicLocationCreateEvent geographicLocationAttributeValueChangeEvent geographicLocationDeleteEvent

TMF642	Alarm Management API	alarmCreateEvent alarmAttributeValueChangeEvent alarmStateChangeEvent alarmDeleteEvent ackAlarmsCreateEvent ackAlarmsStateChangeEvent unAckAlarmsCreateEvent unAckAlarmsStateChangeEvent clearAlarmsCreateEvent clearAlarmsStateChangeEvent commentAlarmsCreateEvent commentAlarmsStateChangeEvent groupAlarmsCreateEvent groupAlarmsStateChangeEvent unGroupAlarmsCreateEvent unGroupAlarmsStateChangeEvent
TMF656	Service Problem Management API	problemAcknowledgementCreateEvent problemAcknowledgementDeleteEvent problemAcknowledgementStateChangeEvent problemGroupCreateEvent problemGroupDeleteEvent problemGroupStateChangeEvent problemUnacknowledgementCreateEvent problemUnacknowledgementDeleteEvent problemUnacknowledgementStateChangeEvent problemUngroupCreateEvent problemUngroupDeleteEvent problemUngroupStateChangeEvent serviceProblemAttributeValueChangeEvent serviceProblemCreateEvent serviceProblemDeleteEvent serviceProblemInformationRequiredEvent serviceProblemStateChangeEvent
TMF621	Trouble Ticket Management API	troubleTicketCreateEvent troubleTicketAttributeValueChangeEvent troubleTicketStatusChangeEvent troubleTicketDeleteEvent troubleTicketResolvedEvent troubleTicketInformationRequiredEvent
TMF632	Party Management API	individualDeleteEvent organizationDeleteEvent
TMF669	Party Role Management API	partyRoleDeleteEvent

TMF662	Entity Catalog Management API	entityCatalogCreateEvent entityCatalogDeleteEvent entityCatalogItemCreateEvent entityCatalogItemDeleteEvent entitySpecificationCreateEvent entitySpecificationDeleteEvent associationSpecificationCreateEvent associationSpecificationDeleteEvent associationCreateEvent associationDeleteEvent
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4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v25.0

SID	v25.0
Functional Framework	v25.0

5.2. Jira References

SID [ISA-1288] Missing Network Trouble Ticket Entity - TM Forum JIRA

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	11-Feb-2025	Yuval Stein	First Version
1.0.0	19-May-2025	Rosie Wilson	Final Administrative Updates
1.0.1	09-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025. Final Administrative Updates
1.1.0	05-Jun-2026	Hugo Vaughan	Gen 5 API updates
1.1.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	19-May-2025	Rosie Wilson	First Version
Pre-production	23-Jun-2025	Adrienne Walcott	Updated to Member Evaluated status

Pre-production	09-Jul-2025	Rosie Wilson	New release 1.0.1
Production	18-Aug-2025	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	25-Jun-2026	Sannah Sibaya	New Release 1.1.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team.

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